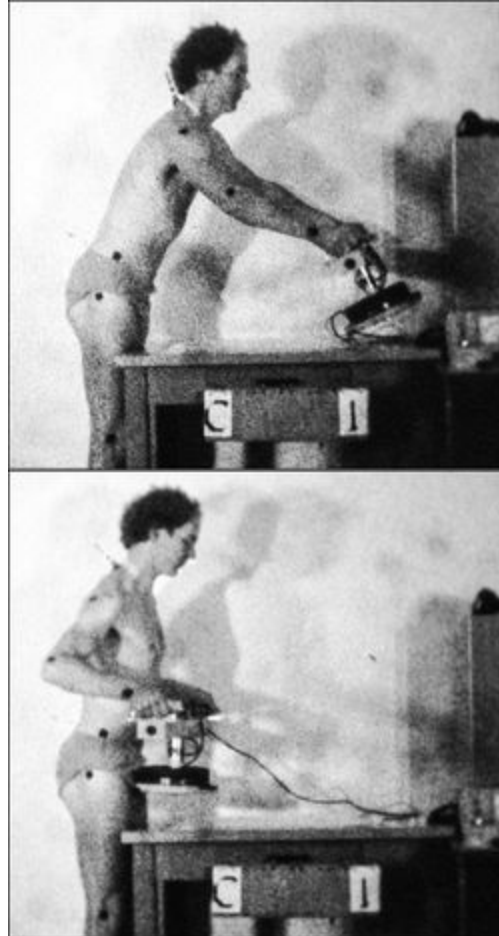


must be stressed that such lifts are conducted by highly practiced and skilled people. In most cases, acceleration of loads to decrease low back stress in the manner described is not suitable for the average person conducting the lifting chores of daily living. The momentum-transfer technique is a skilled movement that requires practice; it is feasible only for awkwardly placed lighter loads and cannot be justified for heavy lifts.

**Figure 7.22** Lifting an awkwardly placed load slowly and smoothly, as common wisdom suggests, compromises the back. Rather, accelerating the torso first and then transferring the momentum to the load as the arms straighten can reduce the spine load. Accelerating the torso and the load at the same time is poor technique that violates this principle and causes higher spine loading. These photographs are from an early experiment in the author's research career.



Photos from Stuart McGill

[Figure 7.23](#) shows another example of the skillful use of this principle by a worker performing the saddle toss lift.

**Figure 7.23** This lift is termed the saddle toss by some; the load is swung with knee contact, minimizing muscular force in the back.



Photos from Stuart McGill

Another mechanical variable should be integrated into the analysis of a dynamic technique. The tissue property of viscoelasticity enables tissues to sustain higher loads when loaded quickly (Burstein and Frankel, 1968). Troup (1977) suggested that viscoelasticity may be used to increase the margin of safety for spine injury during a higher strain rate but cautioned that incorporating this principle into lifting technique should depend on the rate of increase in spinal stress, the magnitude of peak stress, and its duration. Tissue viscoelasticity means that under faster loading, the tissues do not have time to deform, even when the magnitude of the force is high. In this way the critical levels of tissue deformation required to cause damage are not reached. But

given variability in response to load rate among the tissues, and among individuals, no specific guidance pertaining to actual load rate can be offered here.

The instruction to always lift a load smoothly may not invariably result in the least risk of injury. Indeed, it is possible to skillfully transfer momentum to an awkwardly placed object to position the load close to the body quickly and minimize the extensor torque required to support the load. In addition, tissue viscoelasticity can be protective during higher load rates. Clearly, reducing the extensor moment required to support the hand load is paramount in reducing the risk of injury; the best way to accomplish this is to keep the load as close to the body as possible.

## **Is There Any Way to Make Seated Work Less Demanding on the Back?**

Prolonged sitting is problematic for the back. Unfortunately, this fact seems to be rather unknown in the occupational world. Those recovering from back injuries who return to modified work are often given light duties that involve prolonged sitting. Such duties are perceived as being easy on the back, but this can be far from the truth. Even though the returning worker states that she cannot tolerate sitting, that in fact she would be more comfortable walking and even lifting, she is accused of malingering. This is the result of a misunderstanding of sitting mechanics.

### **Sitting Studies**

Epidemiological evidence presented by Videman, Nurminen, and Troup (1990) documented the increased risk of disc herniation in those who perform sedentary jobs characterized by sitting. Known mechanical changes associated with the seated posture include the following:

- Increase in intradiscal pressure when compared to standing postures (Nachemson, 1966)
- Increase in posterior annulus strain (Pope et al., 1977)
- Creep in posterior passive tissues (McGill and Brown, 1992), which decreases anteroposterior stiffness and increases shearing movement

(Schultz et al., 1979)

- Posterior migration of the mechanical fulcrum (Wilder, Pope, and Frymoyer, 1988), which reduces the mechanical advantage of the extensor musculature (resulting in increased compressive loading)

These changes caused by prolonged sitting have motivated occupational biomechanists attempting to reduce the risk of injury to consider the duration of sitting as a risk factor when designing seated work. A recently proposed guideline suggested a sitting limit of 50 minutes without a break, although this proposal will be tested and evaluated in the future.

## **Strategies to Reduce Back Troubles During Prolonged Sitting**

We have developed a four-point approach for reducing back troubles associated with prolonged sitting:

1. Use an ergonomic chair, but use it properly (very few actually do). Many people believe that they should adjust their chair to create the ideal sitting posture. Typically, they adjust the chair so that the hips and knees are bent to 90° and the torso is upright (see [figure 7.24](#)). In fact, this is often shown as the ideal posture in many ergonomic texts. This may be the ideal sitting posture, but for no longer than 10 minutes! Tissue loads must be migrated from tissue to tissue to minimize the risk of any single tissue's accumulating microtrauma. This is accomplished by changing posture. Thus, an ergonomic chair is one that facilitates easy posture changes over a variety of joint angles (see [figure 7.25](#)). Callaghan and McGill (2001a) documented the range of spine postures that people typically adopt to avoid fatigue. Some have three or four preferred angles. The primary recommendation is to continually change the settings on the chair. Many workers continue to believe that there is a single best posture for sitting and are reluctant to try others. This is, of course, unfortunate, because the ideal sitting posture is a variable one. Many employees need to be educated about how to change their chairs and about the variety of postures that are possible.
2. Get out of the chair. There simply is no substitute for getting out of the chair. Some guidelines suggest performing exercise breaks while seated, and some even go as far as to suggest flexing the torso in a stretch. This

is both nonsense and disastrous! A rest break must consist of the opposite activity to reduce the imposed stressors. Extension relieves posterior annulus stress, but more flexion while seated increases it. The recommended break that we have developed involves standing from the chair and maintaining a relaxed standing posture for 10 to 20 seconds. At this stage, some may choose to perform neck rolls and arm windmills to relieve neck and shoulder discomfort from their desk work. The main objective is to buy some time to allow the redistribution of the nucleus and reduce annular stresses. The person then raises the arms over the head (see [figure 7.26](#), a-c) and then pushes the hands upward to the ceiling. By inhaling deeply, the person finds that the low back is fully extended. In this way, the person has taken the back through gentle and progressive lumbar extension without having been taught lumbar position awareness or even understanding the concept. Some will argue that in their jobs they cannot stand and take a break; they must continue their seated work. These people generally will need to be shown the opportunities for standing. For example, they could choose to stand when the phone rings and speak standing. With these simple examples, they will soon see the opportunities to practice this part of good spine health.

3. Remove wallet from back pocket. Viggiani and colleagues (2012) documented the point load in the hip capsule and sciatic nerve when sitting on a wallet (see [figure 7.27](#)). Interestingly, some experiments have used this as an intentional inducer of back pain!
4. Perform an exercise routine at some time in the workday. Midday would be ideal, but first thing in the morning is unwise (see the previous guideline). A good general back routine is presented in the last section of this book.

**Figure 7.24** The so-called ideal sitting posture (90° angles at the hips, knees, and elbows) described in most ergonomic guides. This is erroneous; the ideal sitting posture is one that involves a variety of postures.



Photos from Stuart McGill

**Figure 7.25** Good posture for prolonged sitting is a variable one that migrates the internal loads among the tissues. Possible short-term sitting posture options are shown.





Photos from Stuart McGill

**Figure 7.26** A strategy to thwart the accumulation of disc stresses from (a) prolonged sitting is to (b) stand (note the forward antalgic posture often observed after sitting), (c) reach for the ceiling, stretch pushing the hands upward, inhale deeply and (d) then perform the final extension. This sequence, performed slowly, causes a gentle and progressive extension of the lumbar region and dispels the stresses of sitting.





**Figure 7.27** Map of the pressure distribution between the chair and a person's pelvis and legs showing the concentrations due to the wallet in the back pocket.

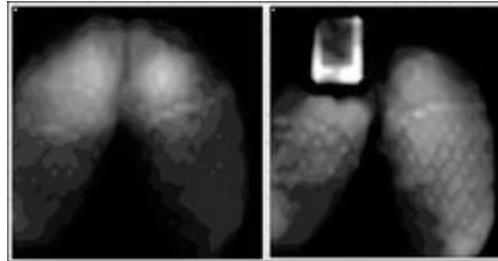
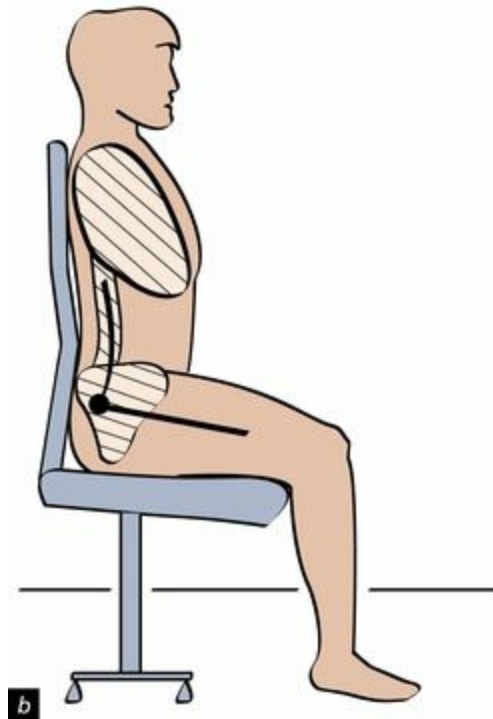


Photo from Daniel Viggiani

A couple of athletic examples may provide insight as well. Athletes who sit on the bench between plays ([figure 7.28a](#)) are often ill prepared for immediately resuming play (Green, Grevier, and McGill, 2002). To help combat this problem, they should sit in taller chairs with angulated seat pans to reduce lumbar flexion ([figure 7.28b](#)), and stand and sometimes pace approximately every 20 minutes. In addition, we question the many exercises performed in a seated position, which appears to be for convenience rather than related to any scientific rationale.

**Figure 7.28** (a) Sitting on the bench with a flexed lumbar spine is problematic because it creates or exacerbates a posterior disc bulge (or both), and it causes loss of the benefits obtained from warming up the back. (b) Sitting in taller chairs with angled seat pans helps extend the hips and lumbar spine.



## Some Short-Answer Questions

The following questions and their answers provide further guidance to reduce the risk of occupational LBD.

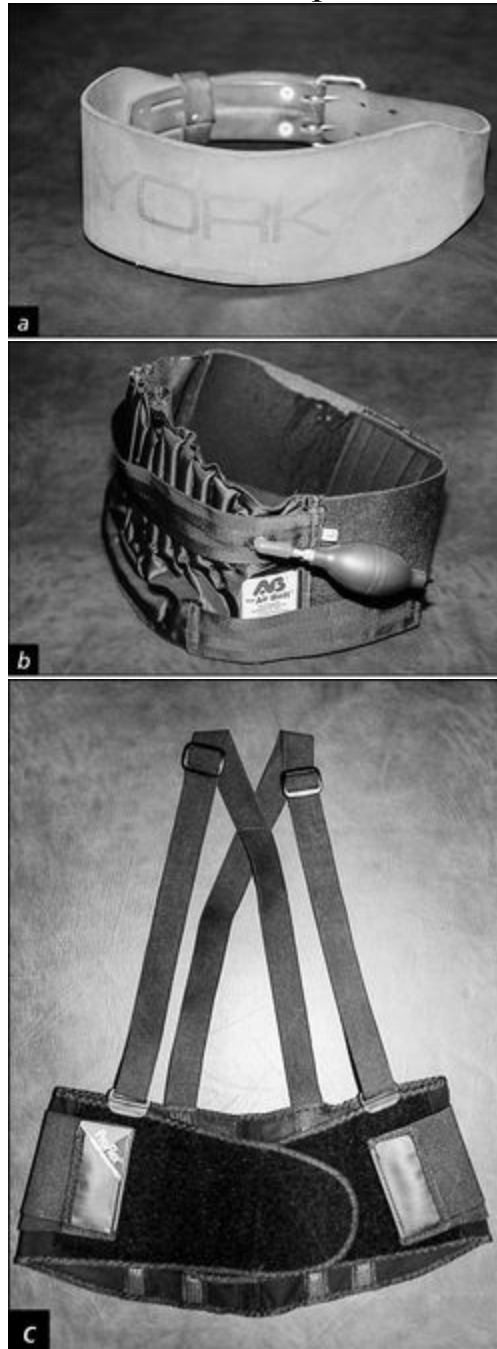
- Is it advisable to maintain a reasonable level of fitness? As much as we would all like to believe that higher levels of fitness are protective for low back disorders, it is argued by some that the literature is not strongly supportive. This is for several reasons. Many clinical trials in which the intervention was designed to enhance fitness actually chose exercises that inadvertently increased the risk of spine damage. For example, many assumed that enhancing abdominal strength was a good idea and addressed this goal by prescribing sit-up exercises. Sit-ups damage the backs of most people; they do not increase back health. Perhaps this has acted as an artifact biasing the literature. Interestingly, the most recent studies that have used biomechanical evidence to develop exercises—particularly stabilization exercises—have shown them to be efficacious. In fact, a review of preventive interventions by Linton and van Tulder (2001) suggests that well-chosen exercise is the most powerful strategy for preventing occupational back disorders. A stable spine maintained with healthy and wise motor patterns, along with higher muscle endurance, is protective.
- Move well. Employing the spine-sparing principles noted here such as hip hinging rather than spine bending has reduced the incidence of back pain in large test groups (e.g., the military; Suni et al., 2013). Choose the most appropriate movement tools (chapter 8) to accomplish the task (e.g., the lunge to move to the floor).
- Should people lift or perform extreme torso bending shortly after rising from bed? The answer is no. Recall the biomechanics of daily changes in the spine as discussed in chapter 4: the discs are hydrophilic and imbibe fluid overnight. This is why people are taller in the morning than when they retire at night. This is also why it is much more difficult to bend in the morning to put one's socks on compared to taking them off at night; the bending stresses are much higher, as is the risk of disc damage. This diurnal variation in spine length and the ability to flex forward have been well documented. As previously noted, Snook and colleagues (1998) found the strategy of restricting forward spine flexion in the morning to be very effective in reducing symptoms in a group of patients with back trouble.

- Should workers adopt a lifting strategy to recruit the lumbodorsal fascia or involve the hydraulic amplifier mechanism? As noted in chapter 4, these mechanisms have been shown to be untenable for sparing the back. Although some still attempt to train workers to invoke these strategies, they have little scientific support. In many cases such strategies are detrimental.
- Should the trunk musculature be cocontracted to stabilize the spine? As noted in chapter 5, the answer is generally yes. In chapter 4, we noted that although such coactivation imposes some penalty on the spine, it is best for the spine to pay this price to enhance stiffness and resist buckling and to reduce the risk of other unstable behavior. How much cocontraction is necessary? As noted in chapter 5, in most tasks, sufficient stability is ensured with very modest amounts of cocontraction—somewhere in the magnitude of 5 to 10% of maximal voluntary contraction for the abdominal wall and other antagonists. Achieving added stiffness in the spine through cocontraction will help prevent injuries that can otherwise result even while lifting a very light object, sneezing, or tying one's shoe. Of course, when the potential exists for surprise loading or handling nonrigid material or heavier loads, the coactivation magnitudes must increase. Damaged joint tissues also require higher levels of cocontraction to avoid unstable joint behavior. The necessary levels of coactivation would depend specifically on the task and the person's history. Thus, lightly cocontracting the stabilizing musculature upon exertion is a reasonable guideline; this should become automatic following stability training for those who do not naturally stiffen.

## **The Question of Back Belts**

This section focuses on occupational belt use (see [figure 7.29](#), a-c, for the typical types of belts used and tested). For information on the athletic use of belts, see *Ultimate Back Fitness and Performance* (McGill, 2014) or McGill (2008). After reading this section, you will be able to make decisions on who should wear a belt and to justify their prescription and use.

**Figure 7.29** Several types of belts are worn and have been tested; (a) the leather belt, (b) the inflatable cell belt, and (c) the stretch belt with suspenders are a few examples.



Photos from Stuart McGill

## Issues of the Back Belt Question

The average patient must be confused when observing both Olympic athletes and people with back injuries wearing abdominal belts. Years ago I conducted a review of the effects of belt wearing (McGill, 1993) and summarized my findings as follows:

- Those who have never had a back injury appear to receive no additional protective benefit from wearing a belt.
- It would appear that those injured while wearing a belt risk a more severe injury than if they had not worn a belt.
- Belts appear to instill in people the belief that they can lift more and may in fact enable them to lift more.
- Belts appear to increase intra-abdominal pressure and blood pressure.
- Belts appear to change the lifting styles of some people to either decrease or increase spine load.

As more studies became available, I updated the review (McGill, 1999a, 1999b). Even though there were more data, the conclusion did not change then and has not changed since. In summary, given the assets and liabilities of belt wearing, I do not recommend them for healthy people either in routine work or in exercise participation. However, the temporary prescription of belts may help some workers return to work.

Manufacturers of abdominal belts and lumbar supports continue to sell them to industry in the absence of a regulatory requirement to conduct controlled clinical trials similar to those required for drugs and other medical devices. Many claims have been made about how abdominal belts reduce injury. For example, some have suggested that belts do the following:

- Remind people to lift properly.
- Support shear loading on the spine that results from the effect of gravity acting on the handheld load and mass of the upper body when the trunk is flexed.
- Reduce compressive loading of the lumbar spine through the hydraulic action of increased intra-abdominal pressure associated with belt wearing.
- Act as a splint, reducing the range of motion and thereby decreasing the risk of injury.
- Provide warmth to the lumbar region.

- Enhance proprioception via pressure to increase the perception of stability.
- Reduce muscular fatigue.

These ideas, among others, are addressed in this chapter. The following section addresses the science regarding the occupational use of belts and concludes with evidence-based guidelines.

The 1994 U.S. National Institute for Occupational Safety and Health (NIOSH) report, “Workplace Use of Back Belts,” contained critical reviews of a substantial number of scientific reports evaluating back belts. The report concluded that back belts do not prevent injuries in uninjured workers nor are they protective for those who have not been injured. Although this is generally consistent with our position stated in 1993, my personal position on belt prescription is somewhat more moderate.

## **Scientific Studies**

In this section I divide the scientific studies into clinical trials and those that examined biomechanical, psychophysical, and physiological changes from belt wearing. Finally, based on the evidence, I recommend guidelines for the prescription and use of belts in industry.

### **Clinical Trials**

Many clinical trials reported in the literature were fraught with methodological problems and suffered from the absence of a matched control group, no posttrial follow-up, limited trial duration, and insufficient sample size. As a result, I review only a few clinical trials in this chapter, while acknowledging the extreme difficulty in executing such trials.

- Walsh and Schwartz (1990) divided 81 male warehouse workers into three groups:
  - A control group (n = 27)
  - A group that received a half-hour training session on lifting mechanics (n = 27)
  - A group that received a 1-hour training session and wore low back

orthoses while at work for the subsequent six months (n = 27)

Instead of using more common types of abdominal belts, this research group used orthoses with hard plates that were heat molded to the low back region of each worker. Given the concern that belt wearing was hypothesized to cause the abdominal muscles to weaken, the researchers measured the abdominal flexion strength of the workers both before and after the clinical trial. The control group and the training-only group showed no changes in abdominal flexor strength or any change in lost time from work. The third group, which received training and wore the belts, showed no changes in abdominal flexor strength or accident rate but did show a decrease in lost time. However, the increased benefit appeared only to accrue to workers who had had a previous low back injury. Van Poppel and colleagues (1998) reached a similar conclusion in a study of 312 airline baggage handlers.

- Reddell and colleagues (1992) studied 642 baggage handlers who worked for a major airline. They divided the baggage handlers into four treatment groups:
  - A control group (n = 248)
  - A group that received only a belt (n = 57)
  - A group that received only a 1-hour back education session (n = 122)
  - A group that received both a belt and a 1-hour education session (n = 57)

The trial lasted 8 months, and the belt used was a fabric weightlifting belt 15 cm (6 in.) wide posteriorly and approximately 10 cm (4 in.) wide anteriorly. The researchers noted no significant differences among treatment groups for total lumbar injury incident rate, lost workdays, or workers' compensation rates. Although the lack of compliance by a significant number of subjects in the experimental group was cause for consideration, those who began wearing belts but discontinued their use had a higher lost-day case injury incident rate. In fact, 58% of workers belonging to the belt-wearing groups discontinued wearing belts before the end of the 8-month trial. Further, an increase in the number and severity of lumbar injuries occurred following the trial in those who previously wore belts.

- Mitchell and colleagues (1994) conducted a retrospective study of 1,316 workers who performed lifting activities in the military. Although this

study relied on self-reported physical exposure and injury data over 6 years prior to the study, the authors did note that the costs of back injuries that occurred while wearing a belt were substantially higher than the costs of injuries sustained while not wearing a belt.

- Kraus and colleagues (1996), in a widely reported study, surveyed the low back injury rates of nearly 36,000 employees of the Home Depot stores in California from 1989 to 1994. During this study period, the company implemented a mandatory back belt use policy. Injury rates were recorded. Even though the authors claimed that belt wearing reduced the incidence of low back injury, an analysis of the data and methodology suggested that a much more cautious interpretation may be warranted. The data showed that whereas belt wearing reduced the risk in younger males and those older than 55, belt wearing appeared to increase the risk of low back injury in men working longer than 4 years by 27% (although the large confidence interval required an even larger increase for statistical significance) and in men working less than 1 year. However, of greatest concern is the lack of scientific control to ferret out the true belt-wearing effect: there was no comparable non-belt-wearing group, which is critical given that the belt-wearing policy was not the sole intervention at Home Depot. For example, over the period of the study, the company increased the use of pallets and forklifts, installed mats for cashiers, implemented postaccident drug testing, and enhanced worker training. In fact, a conscious attempt was made to enhance safety in the corporate culture. This was a large study, and the authors deserve credit for the massive data reduction and logistics work. However, despite the title and claims that back belts reduce low back injury, this uncontrolled study cannot answer the question about the effectiveness of belts.
- Wassell and colleagues (2000), in response to the huge promotion of the Kraus and colleagues (1996) paper by some interest groups and the study's methodological concerns, replicated Kraus and colleagues' work under the sponsorship of NIOSH. These researchers, however, used better scientific control to evaluate only the effect of the belts. Surveying over 13,873 employees at newly opened stores of a major retailer, in which 89 stores required employees to wear belts and 71 stores had only voluntary use, they discovered that the belts failed to reduce the incidence of back injury claims. This study has more power than the

Kraus study.

- Calmels and colleagues (2009) provided people with recent-onset back pain with belts with metal stays, which fall into the brace, or corset, category. Subjects showed a very modest reduction in pain scales, on average, over the subsequent 90 days. The standard deviation was larger than the main effect, suggesting that the belt really helped some and not others. This is consistent with past studies.

In summary, difficulties in executing a clinical trial are acknowledged. The Hawthorne effect is a concern, because it is difficult to present a true double-blind paradigm to workers (those who receive belts certainly know so). In addition, logistical constraints on duration, diversity in occupations, and sample size are problematic. However, the data reported in the better-executed clinical trials cannot support the notion of the universal prescription of belts to all workers involved in the manual handling of materials to reduce the risk of low back injury. Weak evidence suggests that those already injured may benefit from belts (or molded orthoses) with a reduced risk of injury recurrence. However, evidence does not appear to support uninjured workers' wearing belts to reduce the risk of injury; in fact, the risk of injury seems to increase following a trial of belt wearing. Finally, some evidence suggests that the cost of a back injury may be higher in workers who wear belts than in workers who do not.

## **Biomechanical Studies**

Researchers who have studied the biomechanical issues of belt wearing have focused on spinal forces, intra-abdominal pressure (IAP), load, and range of motion. The most informative studies are reviewed in this section.

### **IAP and Low Back Compressive Load**

Biomechanical studies have examined changes in low back kinematics and posture in addition to issues of specific tissue loading. Two studies in particular (Harman et al., 1989, and Lander, Hundley, and Simonton, 1992) suggested that wearing an abdominal belt can increase the margin of safety during repetitive lifting. Both of these papers reported ground reaction force and measured IAP while subjects repeatedly lifted barbells. Both reports

observed an increase in IAP in subjects who wore abdominal belts. These researchers assumed that IAP is a good indicator of spinal forces, which is highly contentious. Nonetheless, they assumed that the higher recordings of IAP indicated an increase in low back support that in their view justified the use of belts. Neither study measured or calculated spinal loads.

Several studies have questioned the hypothesized link between elevated IAP and reduction in low back load. For example, using an analytical model and data collected from three subjects lifting various magnitudes of loads, McGill and Norman (1987) noted that a buildup of IAP required additional activation of the musculature in the abdominal wall, resulting in a net increase in low back compressive load and not a net reduction of load, as researchers had previously thought. In addition, Nachemson, Andersson, and Schultz (1986) published some experimental results of directly measured intradiscal pressure during the performance of Valsalva maneuvers, documenting that an increase in IAP increased, not decreased, the low back compressive load. Therefore, the conclusion that an increase in IAP due to belt wearing reduces compressive load on the spine seems erroneous. In fact, such an increase may have no effect or may even increase the load on the spine.

#### **IAP and Low Back Muscles**

Several studies have put to rest the belief that IAP affects low back extensor activity. McGill and colleagues (1990) examined IAP and myoelectric activity in trunk musculature while six male subjects performed various types of lifts either wearing or not wearing an abdominal belt (a stretch belt with lumbar support stays, Velcro tabs for cinching, and suspenders for when subjects were not lifting). Wearing the belt increased IAP by approximately 20%. Further, the authors hypothesized that if belts were able to support some of the low back extensor moment, one would expect to measure a reduction in extensor muscle activity. There was no change in activation levels of the low back extensors or in any of the abdominal muscles (rectus abdominis or obliques).

In a study of the effect of belts on muscle function, Reyna and colleagues (1995) examined 22 subjects for isometric low back extensor strength. They found that belts provided no enhancement of function (although this study was only a 4-day trial and did not examine the effects over a longer duration).

Over a 4-week period, Ciriello and Snook (1995) examined 13 men lifting 29 metric tons in 4 hours twice a week both with and without a belt. Median frequencies of the low back electromyographic signal (which is sensitive to local muscle fatigue) were not modified by the presence or absence of a back belt, strengthening the notion that belts do not significantly alleviate the loading of back extensor muscles. Once again, this trial was not conducted over a very long period of time.

### **Belts and Range of Motion Restriction**

In 1986 Lantz and Schultz observed the kinematic range of lumbar motions in subjects wearing low back orthoses. Although they studied corsets and braces rather than abdominal belts, they did report restrictions in the range of motion, although the restricted motion was minimal in the flexion plane.

In another study McGill, Seguin, and Bennett (1994) tested the flexibility and stiffness of the lumbar torsos of 20 male and 15 female adult subjects, both while they wore and did not wear a 10 cm (4 in.) leather abdominal belt. The stiffness of the torso was significantly increased about the lateral bend and axial twist axes when subjects wore belts but not when subjects were rotated into full flexion. Thus, these studies seem to indicate that abdominal belts help restrict the range of motion about the lateral bend and axial twist axes, but do not have the same effect when the torso is forced in flexion, as in an industrial lifting situation.

Posture of the lumbar spine is an important issue in injury prevention for several reasons. In particular, Adams and Hutton (1988) showed that the compressive strength of the lumbar spine decreases when people approach the end range of motion in flexion. Therefore, if belts restrict the end range of motion, one would expect the risk of injury to be correspondingly decreased. Although the splinting and stiffening action of belts occurs about the lateral bend and axial twist axes, stiffening about the flexion–extension axis appears to be less.

A more recent data set presented by Granata, Marras, and Davis (1997) supports the notion that some belt styles are better at stiffening the torso—namely, the taller elastic belts that span the pelvis to the rib cage. Furthermore, these authors also documented that a rigid orthopedic belt generally increased the lifting moment, whereas the elastic belt generally

reduced spinal load. Nevertheless, the authors noted a wide variety in subject response. (Some subjects experienced increased spinal loading with the elastic belt.) Even in well-controlled studies, belts appear to modulate lifting mechanics in some positive ways in some people and in negative ways in others.

### **Studies of Belts, Heart Rate, and Blood Pressure**

Hunter and colleagues (1989) monitored the blood pressure and heart rate of five males and one female performing deadlifts and one-arm bench presses and riding bicycles while wearing and not wearing a 10 cm (4 in.) weight belt. Subjects were required to hold in a lifting posture a load of 40% of their maximal weight in the deadlift for 2 minutes. The subjects were required to breathe throughout the duration so that no Valsalva effect occurred. During the lifting exercise, blood pressure (up to 15 mmHg) and heart rate were both significantly higher in subjects wearing belts. Given the relationship between elevated systolic blood pressure and an increased risk of stroke, Hunter and colleagues (1989) concluded that people who may have cardiovascular system compromise are probably at greater risk when undertaking exercise while wearing back supports than when not wearing them.

Subsequent work conducted in our own laboratory (Rafacz and McGill, 1996) involved the investigation of the blood pressure of 20 young men performing sedentary and very mild activities both with and without a belt. (The belt was the elastic type with suspenders and Velcro tabs for cinching at the front.) Wearing this type of industrial back belt significantly increased diastolic blood pressure during quiet sitting and standing both with and without a handheld weight, during a trunk rotation task, and during a squat lifting task. Evidence increasingly suggests that belts increase blood pressure!

Over the past decade I have been asked to deliver lectures and participate in academic debates on the back belt issue. On several occasions occupational medicine personnel have approached me after hearing of the effects of belts on blood pressure and IAP and have expressed suspicions that long-term belt wearing at their particular workplace may possibly be linked with more incidents of varicose veins in the testicles, hemorrhoids, and hernias. As of this writing, there has been no scientific and systematic investigation of the validity of these suggestions. Rather than wait for strong scientific data to either lend support to these ideas or dismiss them, it may be

prudent to simply state concern. This will motivate studies in the future to track the incidence and prevalence of these pressure-related disorders to assess whether they are indeed linked to belt wearing.

Additional studies have examined the effect of belt wearing on other physiological phenomena. Bobick and colleagues (2001) reported lower mean oxygen consumption, and Parker and colleagues (2000) reported reductions in lung ventilation tidal volumes in groups with higher abdominal fat, but increased tidal volumes in groups with lower fat. Collectively, these data would have implications regarding lower work rates during belt wearing, or modulated lung function that would affect performance, or both. Once again, the effect of belt wearing appears to depend on the characteristics of the worker, making it difficult to justify a single, universal belt-wearing policy or guideline.

## **Psychophysical Studies**

Some have expressed concern that wearing belts fosters an increased sense of security that may or may not be warranted. Studies based on the psychophysical paradigm allow workers to select weights that they can lift repeatedly using their own subjective perceptions of physical exertion. McCoy and colleagues (1988) examined 12 male college students while they repetitively lifted loads from the floor to knuckle height at the rate of three lifts per minute for 45 minutes. They repeated this lifting bout three times, once without a belt and once each with two types of abdominal belts: a belt with a pump and air bladder posteriorly and the elastic stretch belt described in the McGill, Norman, and Sharratt (1990) study. After examining the magnitudes of loads that subjects had selected to lift in the three conditions, the researchers noted that wearing belts increased the load that subjects were willing to lift by approximately 19%. This evidence may lend some support to the theory that belts give people a false sense of security.

## **Summary of Back Belt Prescription Guidelines**

My earliest report on back belts (McGill, 1993) presented data and evidence that neither completely supported nor condemned the wearing of abdominal belts for industrial workers. After more laboratory studies and

field trials, my position (which has been implemented by several governments and corporations) has not changed.

Given the available literature, it would appear that the universal prescription of belts (i.e., providing belts to all workers in an industrial operation) is not in the best interest of globally reducing both the risk of injury and compensation costs. Uninjured workers do not appear to receive any additional benefit from belt wearing and, in fact, may be exposing themselves to the risk of a more severe injury if they were to become injured. Moreover, they may have to confront the problem of weaning themselves from the belt. However, if some workers perceive a benefit from belt wearing, they should be allowed to wear a belt conditionally but only on trial. The mandatory conditions for prescription (for which there should be no exception) are as follows:

- Given the concerns regarding increased blood pressure and heart rate and issues of liability, all candidates for belt wearing should be screened for cardiovascular risk by medical personnel.
- Given the concern that belt wearing may provide a false sense of security, belt wearers must receive education on lifting mechanics (back school). All too often, belts are promoted to industry as a quick fix to the injury problem. The promotion of belts, conducted in this way, is detrimental to the goal of reducing injury because it redirects the focus from the cause of the injury. Education programs should include information on how tissues become injured, techniques to minimize musculoskeletal loading, and what to do about feelings of discomfort to avoid disabling injury.
- Consultants and clinicians should not prescribe belts until they have conducted a full ergonomic assessment of the person's job. The ergonomic approach should examine and attempt to correct the cause of the musculoskeletal overload and provide solutions to reduce the excessive loads. In this way, belts should only be used as a supplement for a few people, whereas a greater plantwide emphasis should be on the development of a comprehensive ergonomics program.
- Belts should not be considered for long-term use. The objective of any small-scale belt program should be to wean workers from the belts by insisting on mandatory participation in comprehensive fitness programs and education on lifting mechanics, combined with ergonomic

assessment. Furthermore, consultants would be wise to continue their vigilance in monitoring former belt wearers for a period of time following belt wearing, given that this period appears to be characterized by an elevated risk of injury.

## **LBD Prevention for Employers**

Not all prevention strategies can be implemented solely by workers. Employers, too, play a role, which is outlined in the following guidelines.

- Provide protective clothing to facilitate the least stressful postures. Workers sometimes handle material that is too noxious or too hazardous to hold close to their bodies. For example, I have been involved in cases in which workers held dirty material away from the body to keep their shirts clean, unnecessarily loading their backs. The solution was to provide protective coveralls to spare workers' backs. Leather aprons are helpful if the material is sharp to foster holding the load against the abdomen, as for sheet metal workers, for example. Knee pads are good for prolonged work on the ground. Once the employer has provided the necessary protective clothing, workers will figure out the variety of working postures that can spare their backs.
- Should abdominal belts be prescribed to manual materials handlers? Employers are encouraged to follow the prescription guidelines in the previous section of this chapter.
- Optimize containers and packaging of raw material to spare workers' backs. Often in the design of the industrial process, employers don't consider sparing workers' joints. When considering the industrial process, employers should consider whether handled materials can be bulk containerized. Can raw material be handled in smaller bundles or in bundles of different dimensions? Sometimes the matter is as simple as contacting the suppliers to find alternate ways of packaging material that foster handling in a way that conserves the body joints. The purchasing department can play a role in reducing injury! Bins and containers with folding sides help if parts must be picked up out of bins.
- Encourage workers to practice lifting and work, or task, kinematic patterns. Some people simply do not move, bend, and work in ways that spare their spines. In a recent study (McGill et al., 2003), we noted that workers who had had a history of back troubles were more likely to adopt motion patterns that resulted in higher spine loads! Kinematic patterns need to be practiced and grooved into movement repertoires. (Remember the expression quoted earlier: Practice does not make

perfect; it makes permanent.) Some people must practice the spine-conserving motions every day—especially before attempting a particularly strenuous task. Even high-performance athletes must continually regroove motion patterns daily. Some worker groups have attempted to fabricate their own job-specific training and practice equipment. The dummy sitting in a wheelchair in [figure 7.30](#) is an example of this type of worker professionalism. Nurses built the dummy from plumbing pipes and use it to practice one- and two-person patient lifts.

**Figure 7.30** Nurses at a patient care facility fabricated this dummy to practice their patient lifts—an example of worker professionalism.



Photos from Stuart McGill

- Optimize worker rest breaks. A properly designed rest break consists of the opposite activity (and consequently the opposite loading) from that required by the job. For example, the sedentary secretary is best served by a dynamic rest break. The welder, on the other hand, is better served with rest and perhaps a stretch. The following example illustrates a violation of this principle that caused grief. Back in the 1960s, operators in a power plant monitored the process from a chair and had to respond to a vigilance buzzer on their desks that went off every 10 minutes (see [figure 7.31](#), a and b). At each buzzer interval they would stand and walk around the control panel making adjustments. There was no history of back troubles. The room recently became obsolete and was replaced with an updated control room. The design team for the new control room believed that rising from the chair every 10 minutes was too strenuous.

Consequently, the job was redesigned so that the operators were able to stay seated to perform all operations. These operators worked 12-hour shifts. Having the workers sit for this period of time (and removing the need to get out of the chair regularly) resulted in an increase in low back problems. The power plant then hired a consulting group that recommended rest breaks that consisted of having the workers sit on an exercise bike. The workers' backs failed to improve because the consultants did not understand that the rest break must not exacerbate or replicate the forces of the job. In this case, sitting on a bike was not a break from sitting on the job.

**Figure 7.31** (a) The first control room was built in the 1960s and required the operator to stand every 10 minutes to respond to the vigilance buzzer and make an adjustment to the analog instrumentation. There were no reported back troubles. (b) This room was replaced with a new layout based on the designers' assumption that standing every 10 minutes was too strenuous; the operators were able to sit for hours. A huge low back problem emerged.



Photos from Stuart McGill

- Involve workers in the ergonomic process. Design teams often neglect to consult with the expert on a job—the worker who has done it for years. Quite often, the worker knows of a good solution, and it is simply a matter of listening and facilitating the intervention. An added benefit, psychologists claim, is that workers are more likely to comply with the intervention if they perceive themselves to be a major part of the solution.
- Design work to be variable. As several previous examples have documented, accumulation of tissue stress is thwarted by a change in posture, loading, or activity. Human beings were not made to perform repetitive work that emphasizes only a few tissues. Nor were humans

designed not to be stressed. Research has established that tissues adapt and remodel in response to load (e.g., bone: Carter, 1985; ligament: Woo, Gomez, and Akeson, 1985; disc: Porter, 1992; vertebrae: Brinckmann, Biggemann, and Hilweg, 1989). Too little activity can be as problematic as too much. Krismer and colleagues' study (2001) strongly reinforces the idea so frequently stated in this text that the object of good work design is not to make every job easier; in fact, some jobs should be made more demanding for optimal health. In the Krismer study, students who reported low back pain were distinguished from those who did not by several risk factors. Two of those factors were as follows:

- Watching TV or playing computer games more than 2 hours per day (sedentary)
- Regularly going beyond personal limits in sport activities (too much loading)

Good occupational health from a musculoskeletal perspective is achieved when people perform a variety of tasks with well-designed rest activities, along with all of the traditional components such as proper nutrition, adequate sleep, and avoidance of smoking. Design work to be variable!

As illustrated in the previous examples and guidelines, management can play a role in reducing back troubles in workers. It is a mistake to think that management does not need to understand the science to justify specific injury prevention approaches. As we saw previously, injury prevention involves a thorough understanding of the industrial process, the way goods and materials are purchased, protective equipment, appropriate training, the costs and benefits of intervention, and so forth. Training of both workers and management ensures the best results.

## Injury Prevention Primer

To use biomechanics to its full potential in injury prevention, workers and employers must have a reasonable understanding of the concepts as we understand them today. Workers must be educated in the biomechanically justifiable principles described earlier using examples with which they are familiar. Armed with the general principles, they can tackle any job and devise the best joint-conserving strategies. The intention is to enhance the industrial process and enable workers to retire in good health. The highlights of this chapter are summarized here:

- First and foremost, design work and tasks that facilitate variety. Perhaps the most important guideline should be this: Don't do too much of any one thing. Both too much and too little loading are detrimental.
  - Too much of any activity leads to trouble. Relief from cumulative tissue strains is accomplished with posture changes or, better yet, other tasks that have different musculoskeletal demands.
  - Although the tasks of many jobs cannot be changed, the working routines and arrangement of tasks within a job can be designed scientifically to incorporate this principle. Sometimes it is as easy as continually changing the sitting posture.
- Spine power: Keep it low. Power is the product of force and velocity. High power damages discs.
  - If the force on the spine is high, the movement velocity (bending velocity) must be low, ideally zero.
  - If the movement velocity is high, then force must be low. In contrast, the hip joints were designed to generate and sustain power.
- During all loading tasks, avoid a fully flexed or bent spine and rotate the trunk using the hips (preserving a neutral curve in the spine). Doing this has the following benefits:
  - Disc herniation cannot occur.
  - Ligaments cannot be damaged, because they are slack.
  - The anterior shearing effect from ligament involvement is minimized, and the posterior supporting shear of the musculature is maximized.

- Compressive testing of lumbar motion units has shown increases in tolerance with partial flexion but a decreased ability to withstand compressive load at full flexion.
- During lifting, choose a posture to minimize the reaction torque on the low back (either stoop or squat or do something that is somewhere in between), but keep the external load close to the body.
  - A neutral spine is still maintained, but sometimes the load can be brought closer to the spine with bent knees (squat lift) or relatively straight knees (stoop lift). The key is to reduce the torque, which has been shown to be a dominant risk factor.
  - When exerting force with the hands or shoulders, try to direct the line of the force through the low back. This will reduce the reaction torque and the spine load from muscle contraction.
- Consider the transmissible vector. Attempt to direct external forces through the low back to minimize the moment arm, which causes high torques and crushing muscle forces. This principle should be applied when applying pushing and pulling forces while opening a door, when vacuuming, and when performing other household chores.
- Use techniques that minimize the weight of the load being handled. The log-lifting example given in this chapter demonstrates how to lift an entire log into the back of a truck by lifting no more than half of its weight at any point.
- Balance loads between both hands. Carrying a load in one hand creates more spine load than carrying the same load in both hands—and doubling the total weight carried (McGill, Marshall, and Andersen, 2013). (See [figure 4.17](#) in chapter 4.)
- Allow time for the disc nucleus to equilibrate, the ligaments to regain stiffness, and the stress on the annulus to equalize after prolonged flexion (e.g., sitting or stooping), and do not immediately perform strenuous exertions.
  - After prolonged sitting or stooping, spend time standing.
  - This principle can be adapted to many jobs, but some workers do not have the luxury of being able to take the time to allow the disc nucleus to equilibrate. For example, ambulance drivers are often called on to lift heavy loads immediately after significant periods of driving. A strategy for them is to use a lumbar support in the seat while driving to the incident so that the spine is not flexed. Thus, it

can be prepared for the load with minimal disc equilibration (part of the process of warm-up).

- Avoid lifting or spine bending shortly after rising from bed.
  - Forward-bending stresses on the disc and ligaments are higher after rising from bed compared with later in the day (at least 1 or 2 hours after rising), causing discs to become injured at lower levels of load and degree of bending.
  - This principle is problematic for some occupations such as firefighting, in which workers are often aroused from sleep to attend a fire. Such workers should not sit in a slouched posture with the spine flexed when traveling to the scene, but rather, sit upright. In this way the spine will be best prepared for strenuous work without a warm-up.
- Prestress and stabilize the spine even during light tasks.
  - Lightly cocontract the stabilizing musculature to remove the slack from the system and stiffen the spine even during light tasks such as picking up a pencil. The exercises shown in chapter 10 were chosen to groove these motor patterns.
  - Mild cocontraction and the corresponding increase in stability increase the margin of safety of material failure of the column under axial load.
- Avoid twisting and simultaneously generating high twisting torques.
  - Twisting reduces the intrinsic strength of the disc annulus by disabling some of its supporting fibers while increasing the stress in the remaining fibers under load.
  - Because no muscle is designed to produce only axial torque, the collective ability of the muscles to resist axial torque is limited, and they may not be able to protect the spine in certain postures. The additional compressive burden on the spine is substantial for even a low amount of axial torque production.
  - Generating twisting torque with the spine untwisted may not be as problematic, nor is twisting lightly without substantial torque.
- Use momentum when exerting force to reduce the spine load (rather than always lifting slowly and smoothly, which is an ill-founded recommendation for many skilled workers).
  - This is a skill that sometimes needs to be developed.
  - This strategy is dangerous for heavy loads and should not be used

- for lifting them.
  - It is possible that a transfer of momentum from the upper trunk to the load can start moving an awkwardly placed load without undue low back involvement.
- Avoid prolonged sitting.
  - Prolonged sitting is associated with disc herniation.
  - When required to sit for long periods, adjust posture often, stand up at least every 50 minutes, extend the spine, and, if possible, walk for a few minutes.
  - Organize work to break up bouts of prolonged sitting into shorter periods that are better tolerated by the spine.
- Use good rest break strategies. Customize this principle for various job classifications and demands.
  - Workers engaged in sedentary work are best served by frequent, dynamic breaks to reduce tissue stress accumulation.
  - Workers engaged in dynamic work may be better served by longer and more restful breaks.
- Use protective clothing to foster joint-conserving postures. Use coveralls for dirty material handling, heavy aprons for sharp metals, knee pads for working at ground level, and so on.
- Practice joint-conserving kinematic movement patterns. Some workers need to constantly regroove motion patterns such as locking the lumbar spine when lifting and rotating about the hips.
- Maintain a reasonable level of fitness. While it is better to be undertrained than overtrained, a moderate level of fitness is better than poor fitness.
- Combine these guidelines for special situations. For example, some people have difficulty rolling over in bed because of back pain. Nearly everyone with back pain can be taught to manage pain and still accomplish this task by combining a momentum transfer with the minimal twisting guidelines (see [figure 7.32](#), a-d).

**Figure 7.32** Rolling over in bed can be taught to those who maintain that doing so is too painful. The figure illustrates rolling from the left side to the right. (a) While lying on one side, the patient braces the torso so that the spine does not twist in the steps that follow. (b) Then the upper arm and leg are raised together with the lower arm and leg prying off the floor. (c) This is performed quickly enough to generate momentum that will carry the patient through the roll. (d) The patient should now be resting comfortably on the other side.



Photos from Stuart McGill

- In cases of morning stiffness, consider the bed. Some people with spine joints that generate pain when in a particular posture benefit from a mattress that fosters pain-free positioning. Zhang and colleagues (2009) reported that bed characteristics, together with spine loading, were the most predictive elements of those diagnosed with disc herniation. There are no rules to guide mattress selection, although Haex (2004) suggested that those with angular body types generally do better with firm mattresses covered by a soft pillow top. Haex also reported evidence that the amount of standing lumbar lordosis helped with bed selection. For example, they reported that those in Asian cultures, typified with less lordosis, find comfort with covered futons made of wood slats. Europeans, who tend to have more lordosis, found these less comfortable. Memory foam mattresses are very comfortable for some, particularly those who do not change position often. Rolling up and over the shaped foam, however, disturbs those who need to change position more often.
- Suggest the best positions for sexual activity. Primary contact physicians see a number of couples who have long periods of celibacy because of long-lasting back pain following coitus. Physicians have no guidelines to make recommendations. Our pioneering work ranked various positions for spine load (Sidorkewicz and McGill, 2014a, 2014b). In general, we recommend perfecting the hip hinge rather than spine bending, and avoiding a deviated static spine posture.

## A Note for Consultants

Acting as a consultant, I have made many mistakes, some of which motivated the following tips.

- Don't fall into the trap of thinking that you are the expert and know what is best for a worker (unless you have done the job for years yourself). Always consult the worker. Successful workers have developed personal strategies for working that assist them in avoiding fatigue and injury. Their insights are the result of thousands of hours of performing the task, and they can be very perceptive. Try to accommodate them.
- Do not take the instructions for a specific worker verbatim from the preceding Injury Prevention Primer. Instead, explain the relevant biomechanical principle in language and terminology that are familiar to the worker.
- Do not focus only on the most demanding tasks. Given the links among different tasks from a tissue load perspective, you can often obtain better solutions by considering the full complement of exposures. In a similar vein, some consultants tend to focus on a single metric of risk (e.g., low back compression) or rely on only a few simple solutions. The average ergonomist probably does not have the training necessary for achieving the best solutions for low back problems. Perhaps I am biased because in recent years I have been asked to consult only when consultants' poorly conceived ergonomic approaches have failed. I am requested to become involved when the company faces lawsuits or other issues that have raised the stakes. Remember that many solutions are neither simple nor unidimensional, regardless of your training. Use the Injury Prevention Primer as a checklist to evaluate whether better and more comprehensive solutions are available or possible.
- Do not focus exclusively on the musculoskeletal issue. Rather, look for the opportunities that lie in enhancing the industrial process. Any management board will recognize the worth of a consultant who makes the process more efficient, produces a higher-quality product, or reduces injury compensation costs.
- Movement is taught more effectively when it is considered a complex

movement skill. Simply having an expert (i.e., yourself) demonstrate more effective work techniques will usually fail. Coaching the movement and motor patterns in an interactive session leads to performance enhancement and greater safety (see McGill, 2014, for a full overview of this process).

- Never make a recommendation that is not feasible to implement, for monetary or any other reason.
- Finally, consider becoming a job coach—a person with expertise in ergonomics and reducing the external load applied to the body. Job coaches also have expertise in personal body mechanics and movement patterns needed to further reduce tissue loads and enhance load tolerance. Our consulting experience shows that the payback, in terms of injury costs, far outweighs the expense of setting up the job coach approach.

## Reducing the Risk in Athletes

Although this chapter has focused on occupational back injury prevention, these principles are also appropriate for athletic situations. Athletes and teams from a variety of sporting activities—from world-class professionals to amateurs—have sought my advice as a low back injury consultant. In many cases their painful and disabled backs were ending their careers. But as we have seen in preceding chapters, success in dealing with back disorders requires efforts to address both the cause of the troubles and the most appropriate rehabilitative therapy. In many cases, addressing the cause meant that athletes had to change their techniques. Without exception, they had to change the way they trained. Their backs were breaking down for a reason! As loads applied to the body reach world-class levels, technique must be impeccable. Techniques for athletes are well covered in my other textbook, *Ultimate Back Fitness and Performance* (McGill, 2014).

When I am asked to help a team reduce back injuries, I follow an approach I have developed over the years:

- Understand the challenges. Collect data on players and the team. Injury data together with performance data need to be mined for patterns and relationships. These reveal mechanisms that can be addressed. The more detail, the better. For injuries, we want to know details regarding diagnosis, time of the day or season of symptoms, training regimens, training/recovery schedules, peak and taper schedules, and so on. Performance statistics should include injury resilience (games without injury), points and assists scored, and minutes played, among other things.
- Devise a player assessment protocol relevant to the sport. Using the information obtained from the preceding procedure, we have a clear idea of suspected injury mechanisms and imbalances thwarting performance. We combine this information with what is already established about the science of the sport. This creates our baseline characterization of each player.
- Create a monitoring program. Each player is assessed on a schedule—one for in-season and another for out-of-season. The most successful clubs are on a 3-week schedule during the in-season. The testing is

minimally invasive and obtrusive but includes biomechanical, physiological, neurological and psychological elements. This reveals fatigue that leads to injury, player distress causing detriments in play, and so on. The intention is to keep the machine well tuned and injury-free.

- Follow up to assess program efficacy. Here team and player results are compared to those from previous years.

We have assessed the ability of commercial approaches to predict performance and injury resilience. None have lived up to their claims. Athletes have bodies suited to their sports. For example, basketball players are unique in that their height and different body segment proportions do not allow them to pass traditional movement screens. Their long legs simply do not allow a passing grade on a squat. In terms of physiology, explosive players, for example, can only retain their athleticism for a short period. Otherwise, they would rapidly degrade to ordinary performance and higher risk. These special athletes prosper with shorter playing time intervals.

For athletes in individual sports, I assess a game or two and chart their demands. For example, for a mixed martial arts athlete, I would catalog the amount of time spent in an upright posture, the number of fast, explosive strikes, the time spent in isometric control, and so forth. Then I would devise a training program to meet the demands in a spine- and joint-sparing way.

## **A Final Note**

The practitioners most successful in reducing back injury and disability determine the cause and address it prior to prescribing therapeutic interventions. Use this chapter to guide you.